

Name:

Exam Style Questions

Algebraic Proof



Corbettmaths

Equipment needed: Pen

Guidance

1. Read each question carefully before you begin answering it.
2. Check your answers seem right.
3. Always show your workings

Video Tutorial

www.corbettmaths.com/contents

Video 365



Answers and Video Solutions



1. Prove that the sum of three consecutive integers is divisible by 3.



three consecutive integers ; n
 $n+1$
 $n+2$

$$n + (n+1) + (n+2) = 3n+3$$

$$= 3(n+1)$$

$\therefore$ divisible by 3

(3)

2. Prove $(n+6)^2 - (n+2)^2$ is always a multiple of 8



$$(n+6)(n+6) = n^2 + 12n + 36$$

$$(n+2)(n+2) = n^2 + 4n + 4$$

$$n^2 + 12n + 36 - n^2 - 4n - 4$$

$$= 8n + 32$$

$$= 8(n+4)$$

$\therefore$ divisible by 8 and is therefore a multiple of 8

(4)

3. Prove $(n+10)^2 - (n+5)^2$ is always a multiple of 5



$$(n+10)(n+10) = n^2 + 20n + 100$$

$$(n+5)(n+5) = n^2 + 10n + 25$$

sub.

$$\hline 10n + 75$$

$$= 5(2n + 15)$$

$\therefore$ a multiple of 5

(4)

4. Prove the sum of two consecutive odd numbers is even.



two consecutive odd numbers $2n+1$, $2n+3$

$$\text{Sum of } 2n+1 \text{ \& } 2n+3 = 4n+4$$

$$4(n+1)$$

$\therefore$ an even number.

(3)

5. Prove $(2n + 1)(3n - 2) - (6n - 1)(n - 2)$ is always even



$$6n^2 - 4n + 3n - 2 - (6n^2 - 12n - n + 2)$$

$$6n^2 - n - 2 - (6n^2 - 13n + 2)$$

$$12n - 4$$

$$= 4(3n - 1)$$

$\therefore$ even.

(3)

6. Prove that the sum of three consecutive even numbers is always a multiple of 6



Three consecutive even numbers

$$\begin{array}{l} 2n \\ 2n+2 \\ 2n+4 \end{array}$$

Sum

$$2n + (2n+2) + (2n+4)$$

$$= 6n + 6$$

$$= 6(n+1)$$

$\therefore$ multiple of 6.

(3)

7. Prove the sum of four consecutive odd numbers is always a multiple of 8



four consecutive odd numbers ; $2n+1$
 $2n+3$
 $2n+5$
 $2n+7$

sum of

$$\begin{aligned} & (2n+1) + (2n+3) + (2n+5) + (2n+7) \\ &= 8n + 16 \\ &= 8(n+2) \end{aligned}$$

$\therefore$ a multiple of 8.

(4)

8. Prove $(2n+9)^2 - (2n+5)^2$ is always a multiple of 4



$$\begin{aligned} & (2n+9)(2n+9) - (2n+5)(2n+5) \\ &= 4n^2 + 36n + 81 - (4n^2 + 20n + 25) \\ &= 16n + 56 \\ &= 4(4n+14) \end{aligned}$$

$\therefore$ a multiple of 4.

(4)

9. Prove $(n+1)^2 + (n+3)^2 - (n+5)^2 \equiv (n+3)(n-5)$



$$\begin{aligned} & (n+1)(n+1) + (n+3)(n+3) - (n+5)(n+5) \\ &= n^2 + 2n + 1 + (n^2 + 6n + 9) - (n^2 + 10n + 25) \\ &= 2n^2 + 8n + 10 - n^2 - 10n - 25 \\ &= n^2 - 2n - 15 \\ &= (n+3)(n-5) \end{aligned}$$

(4)

10. Prove the product of two even numbers is always even



Even numbers : $2n$ & $2m$

$$\begin{aligned} \text{product : } 2n \times 2m &= 4mn \\ &= 2(2mn) \end{aligned}$$

$\therefore$ always even.

(3)

11. Prove the product of three consecutive odd numbers is odd



Three consecutive odd numbers : $2n+1$
 $2n+3$
 $2n+5$

$$\text{Product : } (2n+1)(2n+3)(2n+5)$$

$$8n^3 + 36n^2 + 46n + 15$$

$$\underbrace{2(4n^3 + 18n^2 + 23n)}_{\text{even}} + 15$$

$$\text{even} + \text{odd} = \text{odd}$$

(3)

12. Prove algebraically that the sum of the squares of two odd integers is always even.



$$2n+1 \quad \text{and} \quad 2k+1$$

$$(2n+1)^2 + (2k+1)^2$$

$$4n^2 + 4n + 1 + 4k^2 + 4k + 1$$

$$= 4n^2 + 4n + 4k^2 + 4k + 2$$

$$= 2(2n^2 + 2n + 2k^2 + 2k + 1)$$

$$\therefore \underline{\text{even}}$$

(4)

13. Prove that when two consecutive integers are squared, that the difference is equal to the sum of the two consecutive integers.



Two consecutive integers : n
 $n+1$

Sum $n+n+1 = 2n+1$

Difference between squares :

$$\begin{aligned} &= (n+1)^2 - n^2 \\ &= n^2 + 2n + 1 - n^2 \\ &= 2n + 1 \end{aligned}$$

$\therefore$ they are equal.

(4)

14. Given x is a ^{positive} integer.



Prove that $(2x+3)^2 - 3x(x+2)$ is a square number.

~~$(2x+3)(2x+3)$~~

$$4x^2 + 6x + 6x + 9 - 3x^2 - 6x$$

$$4x^2 + 12x + 9 - 3x^2 - 6x$$

$$x^2 + 6x + 9$$

$$(x+3)(x+3)$$

$$= (x+3)^2$$

$\therefore$ a square number.

(3)

15. Prove that the sum of the squares of 3 consecutive positive integers is always 1 less than a multiple of 3.



$$n, n+1, n+2$$

$$n^2$$

$$(n+1)^2 = n^2 + 2n + 1$$

$$(n+2)^2 = n^2 + 4n + 4$$

$$n^2 + (n^2 + 2n + 1) + (n^2 + 4n + 4)$$

$$= 3n^2 + 6n + 5$$

$3n^2 + 6n + 6$ is a multiple of 3 since $3(n^2 + 2n + 2)$

$\therefore 3n^2 + 6n + 5$ is 1 less than a multiple of 3.

(4)

16. Prove algebraically that



$$(4n+1)^2 - (2n-1) \text{ is an even number}$$

for all positive integer values of n .

$$(4n+1)(4n+1) - (2n-1)$$

$$= 16n^2 + 8n + 1 - 2n + 1$$

$$= 16n^2 + 6n + 2$$

$$= 2(8n^2 + 3n + 1)$$

$\therefore$ even

(4)

$$(n-6)(n-6)$$

17. Prove that $3n(3n+4) + (n-6)^2$ is positive for all values of n



$$9n^2 + 12n + (n^2 - 12n + 36)$$

$$10n^2 + 36$$

$$n^2 \geq 0$$

$$10n^2 \geq 0$$

$$10n^2 + 36 > 0$$

$\therefore 10n^2 + 36$ is always positive.

(4)

18. The first five terms of a linear sequence are 5, 11, 17, 23, 29 ...



$$6n \quad 6 \quad 12 \quad 18 \quad 24 \quad 30$$

- (a) Find the n th term of the sequence

$$6n - 1$$

(2)

A new sequence is generated by squaring each term of the linear sequence and then adding 5.

- (b) Prove that all terms in the new sequence are divisible by 6.

$$(6n-1)^2 + 5$$

$$(6n-1)(6n-1) + 5$$

$$36n^2 - 12n + 6$$

$$6(6n^2 - 2n + 1) \therefore \text{divisible by 6}$$

(4)

19. Prove that the product of two consecutive even numbers is a multiple of 4.



Two consecutive even numbers : $2n$
 $2n + 2$

product : $2n(2n+2)$
 $4n^2 + 4n$

$$4n(n+1)$$

$\therefore$ a multiple of 4.

(3)

20. Prove that when any odd integer is squared, the result is always one more than a multiple of 8.



odd integer : $2n + 1$

odd integer squared : $(2n+1)(2n+1)$
 $= 4n^2 + 4n + 1$
 $= 4n(n+1) + 1$

since either n or $n+1$ is even,
that means $n(n+1)$ is even.

4 times even is always a multiple of 8.

$\therefore 4n(n+1) + 1$ is one more than
a multiple of 8.

(4)

21. Prove that the product of two odd numbers is always odd.



$$\text{odd numbers : } \begin{matrix} 2k+1 \\ 2m+1 \end{matrix}$$

$$\text{product } (2k+1)(2m+1)$$

$$4km + 2k + 2m + 1$$

$$\underbrace{2(2km + k + m)}_{\text{even}} + 1_{\text{odd}} = \text{odd}$$

(3)

22. Prove algebraically that every term in the sequence $n^2 - 12n + 38$ is positive.



$$(n-b)^2 - 3b + 38$$

$$(n-b)^2 + 2$$

$$\text{since } (n-b)^2 \geq 0$$

$$(n-b)^2 + 2 > 0$$

(3)

23.



Prove that the sum of the cubes of two consecutive odd integers is always a multiple of 4.

$$2n+1 \quad \text{and} \quad 2n+3$$

$$\begin{aligned} & (2n+1)^3 \\ &= (2n+1)(2n+1)(2n+1) \\ &= (4n^2 + 4n + 1)(2n+1) \\ &= 8n^3 + 12n^2 + 6n + 1 \end{aligned}$$

$$\begin{aligned} & (2n+3)^3 \\ &= (2n+3)(2n+3)(2n+3) \\ &= (4n^2 + 12n + 9)(2n+3) \\ &= 8n^3 + 36n^2 + 54n + 27 \end{aligned}$$

$$16n^3 + 48n^2 + 60n + 28$$

$$4(4n^3 + 12n^2 + 15n + 7)$$

$\therefore$ multiple of 4.

(4)